

Solution Requirements

Date	03 October 2026
Team ID	FITBUDDY-TEAM
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Registration	Allow a user to register with the information required by the application.	High
2	Authentication	Allow registered users to log in and access their personalized dashboard.	High
3	Fitness Input	Capture name/user ID, age, weight, fitness goal and workout intensity.	High
4	AI Workout Generation	Generate a structured seven-day workout plan using Gemini.	High
5	Nutrition / Recovery	Generate a concise goal-aligned nutrition or recovery tip.	Medium
6	Feedback Update	Accept natural-language feedback and regenerate the existing plan.	High
7	Persistence	Store user information and original/updated plans in SQLite.	High
8	BMI	Calculate and display BMI from the supplied height/weight information where available.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Acceptance Criteria
1	Usability	Simple forms and readable result pages.	User can move from input to generated plan without unnecessary steps.
2	Performance	Avoid unnecessary repeated AI calls and provide retry/fallback handling.	Application returns a usable response or clear error when AI is unavailable.
3	Security	Keep Gemini API credentials outside source code.	API key supplied through environment configuration.
4	Reliability	Preserve generated plans and support feedback updates.	Original and updated plan data remain available in storage.
5	Maintainability	Separate routes, AI generators, schemas, database and templates.	Each major responsibility has a dedicated module.

S.No	NFR Category	Requirement Description	Acceptance Criteria
6	Portability	Run through a Python virtual environment and Uvicorn.	Application starts with the documented dependencies and server command.